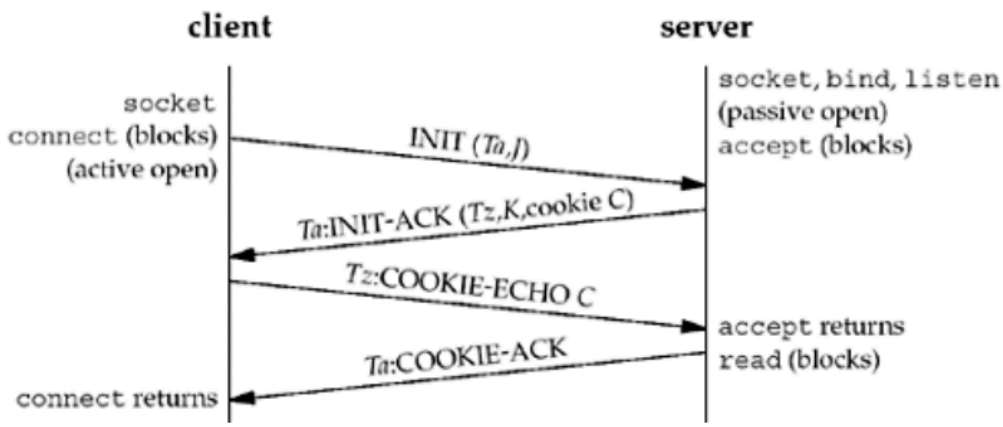
📌 SCTP Association Establishment and Termination - Summary

🔄 Association Establishment (Four-Way Handshake)

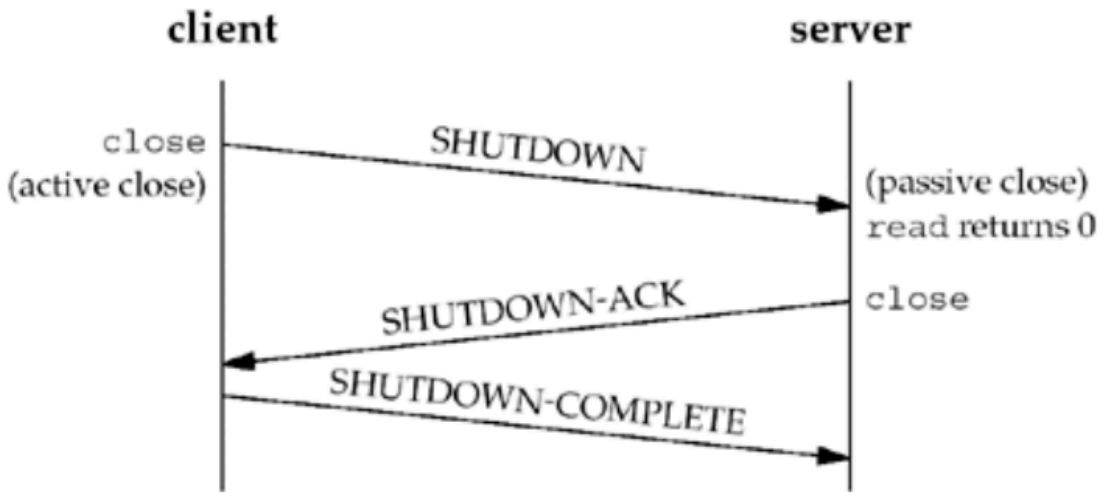
- SCTP is **connection-oriented** like TCP but uses a **four-way handshake** to establish connections (called *associations*).
- **Steps**:
  1. **Passive Open (Server)**: Server prepares using `socket`, `bind`, and `listen`.
  2. **Active Open (Client)**: Client sends an **INIT** message with its IPs, sequence number, and stream info.
  3. **Server Responds**: Sends **INIT-ACK** with its own info and a **state cookie** (digitally signed).
  4. **Client Echoes Cookie**: Sends **COOKIE-ECHO** (may contain data).
  5. **Server Acknowledges**: Sends **COOKIE-ACK** (may also contain data).
- **Why the Cookie?**
  Prevents **denial-of-service (DoS) attacks** by avoiding server-side state creation until the cookie is validated.
- Each side picks a **primary destination IP** for data transfer.
- Tags (`verification tags`) and sequence numbers are used to maintain association integrity.

Figure 2.6. SCTP four-way handshake.



🔚 Association Termination

- SCTP does **not allow half-closed connections** (unlike TCP).
- When one side initiates shutdown:
  - The other side finishes sending queued data.
  - Then both sides complete shutdown.
- No `TIME_WAIT` state is needed—SCTP uses **verification tags** to reject old/duplicate packets.



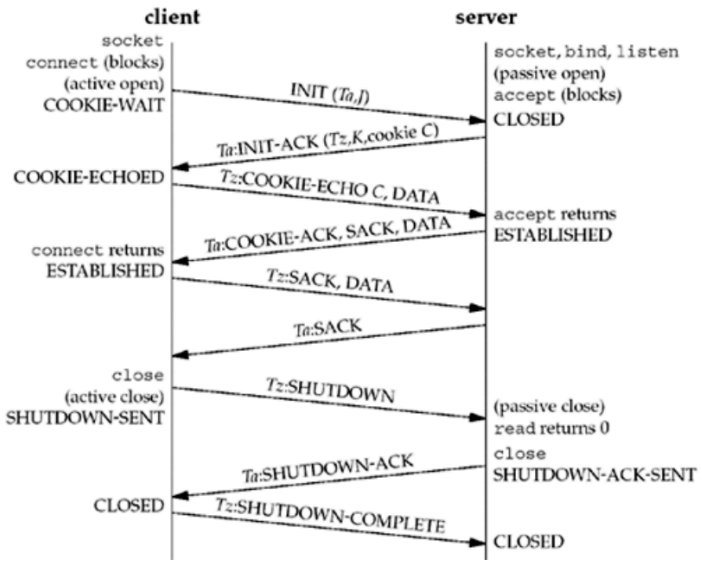
🔁 SCTP State Machine

- States include: `CLOSED`, `COOKIE-WAIT`, `COOKIE-ECHOED`, `ESTABLISHED`, `SHUTDOWN-PENDING`, `SHUTDOWN-RECEIVED`.
- Transitions occur based on received chunk types and actions (e.g., `INIT`, `COOKIE-ACK`, `SHUTDOWN`).

recv: INIT; send INIT-ACK
starting point
CLOSED
appl: active open
send: INIT
COOKIE-WAIT
recv: valid COOKIE ECHO
send: COOKIE ACK
recv: INIT ACK
send: COOKIE ECHO
COOKIE-ECHOED
recv: COOKIE ACK
recv: SHUTDOWN ACK
send: SHUTDOWN COMPLETE
ESTABLISHED
data transfer state
appl: close
send: outstanding DATA
recv: SHUTDOWN
send: outstanding DATA
SHUTDOWN-PENDING
SHUTDOWN-RECEIVED
No more outstanding data
send: SHUTDOWN
No more outstanding data
send: SHUTDOWN ACK
SHUTDOWN-SENT
recv: SHUTDOWN
send: SHUTDOWN ACK
(simultaneous close)
SHUTDOWN-ACK-SENT
recv: SHUTDOWN COMPLETE
recv: SHUTDOWN ACK
send: SHUTDOWN COMPLETE
indicate normal transitions for client
indicate normal transitions for server
appl: indicate state transitions taken when application issues operation
recv: indicate state transitions taken when segment received
send: indicate what is sent for this transition

📦 Packet Exchange Example

- Packets can carry **DATA chunks** alongside control chunks like `COOKIE-ECHO` or `COOKIE-ACK`.
- **Chunks** are the basic units in SCTP packets (self-contained with type, flags, and length).
- Multiple chunks can be bundled in a single SCTP packet.



⚙️ SCTP Extensions and Options

- SCTP supports extensions using **parameters and chunks**.
- Current extensions in development:
    1. **Dynamic Addressing**: Add/remove IPs dynamically in an association.
    2. **Partial Reliability**: Skip sending outdated messages (on application request).